

Lecture notes (2026 Sept. 28)

Slide 1

- Fall break, so no class on Thursday
- No office hours this week
- Homework 5 released: due on October 7

Slide 2

- We can add in constraints for x and u
- If the cost is convex-quadratic and dynamics/constraints are linear, it is a QP

Slide 3

- there might be wind (disturbance)
- we might misjudge our position (state uncertainty)
- we might misjudge where the obstacle is (parameter uncertainty)

Slide 4

- this formulation is general, w_t can be parameters, disturbances, everything in between
- x_t are now all random variables

Slide 5

- the math is a bit more complex, but fun
- Black-Scholes equation

Slide 6

- If there were no disturbances there would be a fixed point
- There is no fixed point

Slide 7

- the stability pushes in, the disturbance pushes out
- We know x_t are Gaussian, so we can consider the mean and covariance

The mean:

$$\begin{aligned}\mathbb{E}x_{t+1} &= \mathbb{E}[A_K x_t] + \mathbb{E}[w_t] \\ &= A_K \mathbb{E}x_t \\ &= A_K^t \mathbb{E}x_0 = 0\end{aligned}$$

The covariance:

$$\begin{aligned}[x_{t+1} x_{t+1}^\top] &= \mathbb{E}[A_K x_t x_t^\top A_K^\top] + \mathbb{E}[A_K x_t w_t^\top] + \mathbb{E}[w_t x_t^\top A_K^\top] + \mathbb{E}[w_t w_t^\top] \\ &= A_K [x_t x_t^\top] A_K^\top + \Sigma = \sum_{i=0}^{t-1} A_K^i \Sigma A_K^{i\top}\end{aligned}$$

which converges to some $\Gamma \succ 0$ because $\rho(A_K) < 1$. Thus,

$$x_t \xrightarrow{d} \mathcal{N}(0, \Gamma)$$

Slide 8

- Algorithm: loop through vertices in $\mathcal{W}$ and vertices in $\mathcal{X}_t$ and make a new set of vertices
- Problem: $O(N^T)$ vertices

Slide 9

- Now, it only takes $O(NT)$ space.

Slide 10

- there is a stationary set here, similar to the SLTI, but it need not be a finite zonotope

Slide 11

Slide 12

Augmented state formulation:

$$\begin{bmatrix} x_{t+1} \\ \xi_{t+1} \end{bmatrix} = \begin{bmatrix} f(x_t, u_t, w_t, \xi) \\ \xi_t \end{bmatrix}, \quad \begin{bmatrix} x_0 \\ \xi_0 \end{bmatrix} \sim \begin{bmatrix} p(x_0) \\ p(\xi) \end{bmatrix}$$

Note: you may lose observability

Slide 13

- Note that J is now a random variable because of stochasticity
- There are things in between average and worst case

Slide 14

- it is hard to do optimization where things are random variables
- sampling is drop-dead easy; but how many do you need?
- does feedback have a role in trajectory optimization? Does it change the number of samples?
- feedback makes the problem non-convex

Slide 15

- Similar feedback story: lack of feedback makes the problem conservative
- the max operation cannot be estimated with sampling

Slide 16

- The idea is to place a tube around the trajectory for the error
- There is feedback K , which could also be a chosen variable.
- We design the tube to always be in the tube, then put the size of the tube as a cost and make the constraints w.r.t. the tube

Slide 17

- If no feedback, the tube blows up.
- Takeaway: open loop in deterministic dynamics is equal with/without feedback, but in stochastic systems that is not the case.

Slide 18

- alpha is a hyperparameter

Slide 19

- The bottom claim comes with an asterisk
- What about just sampling a bunch and ensuring an α failure rate; how many samples would you need?

Slide 20

- we need the independence to replace the value function (last step)
- doesn't need to be identically distributed, but needs to satisfy certain conditions

Slide 21

- similarly, rectangularity is important to decouple timesteps

Slide 22

- note: we satisfy the criteria for the stochastic value function recursion before

Slide 23

- If $\gamma = 1$, we instead need to consider the time-averaged cost-to-go:

$$V(x) = \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E}_w \left[\sum_{t=1}^T c(x_t, u_t, w_t) \right]$$

and recover the original algebraic riccati equation

Slide 24

- note: we can now model POMDPs

Slide 25

- don't ask me to derive these
- very classical in robotics

Slide 26

- iconic abstract
- about robustness margins for LQG vs LQR

Slide 27

- the infinity norm of a hardy space
- Note: idea is that $x_0 = 0$.
- why is there a $\|w(\cdot)\|_{L_2}$ in the denominator
- if linear, it is the same as a unit ball

Slide 28

- follows from squaring and moving things over
- To show, assume that $x_0 = 0$, then:

$$z(t)^\top z(t) \leq \gamma^2 w(t)^\top w(t) - \dot{V}(x(t)).$$

Now integrate from 0 to an arbitrary finite time T :

$$\int_0^T z(t)^\top z(t) dt \leq \gamma^2 \int_0^T w(t)^\top w(t) dt - \int_0^T \dot{V}(x(t)) dt.$$

By the fundamental theorem of calculus,

$$\int_0^T z(t)^\top z(t) dt \leq \gamma^2 \int_0^T w(t)^\top w(t) dt - V(x(T)) + V(x(0)).$$

Because $x(0) = 0$, $V(0) = 0$, and $V(x) \geq 0$:

$$\int_0^T z(t)^\top z(t) dt \leq \gamma^2 \int_0^T w(t)^\top w(t) dt - V(x(T)).$$

$$\implies \|z\|_{L_2[0,T]}^2 \leq \gamma^2 \|w\|_{L_2[0,T]}^2.$$

Slide 29

- whole can of worms
- success of RL is in many ways due to domain randomization

Slide 30

- Note: this formula for CVaR assumes continuous distribution
- CVaR is more risk sensitive than VaR

Slide 31

- A worst case distribution
- D-V lemma and Gibbs distribution
- Wasserstein distance
- Distributionally robust RL

Slide 32

- feedback and uncertainty: helpful, potentially hurtful
- stochastic and robust problems are generally harder to solve
- there are many methods and techniques